

Module IV: Human O&M delivery

3D twin viewer

Analytical dashboard

Grounded NL answers

Gated tickets / revisits



Module II: Twin construction

Geometry

Point clouds

Mesh model

BIM alignment

Deviation check

Perception

Defect detection

Segmentation

Modality tags

Confidence



Twin memory

Observation records

Host attachment

Topology links

Openable evidence



Module III: Language reasoning

Context engine

LLM

RAG retrieval

Answer templates

Citation checks

Tool / write gates

Evidence sources

Twin records

Editioned manuals

Vector index

Neighbour topology

Work-order fields



Module I: Physical acquisition

UAV imagery

Pose / telemetry

BIM / GIS context

Campaign metadata